

GATE PH

Engineering Physics

TREND-BASED MOCK PAPER

— 04 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	9 subjects

Difficulty Distribution

Easy: 14 questions (22%)

Moderate: 42 questions (65%)

Difficult: 9 questions (14%)

GENERAL INSTRUCTIONS

GATE PH — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[General Aptitude]

$5^2 + 12^2$:

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q2. 1M [no -ve]

[Engineering Mathematics]

d/dx of e^x :

Enter your answer here

Q3. 1M [-0.33]

[Classical Mechanics]

$F=ma$ is:

- (A) 1st law
- (B) 2nd law
- (C) 3rd law
- (D) None

Q4. 1M [-0.33]

[Classical Mechanics]

Period of spring-mass:

- (A) $2\pi\sqrt{m/k}$
- (B) $2\pi\sqrt{k/m}$
- (C) m/k
- (D) k/m

Q5. 1M [-0.33]

[Electromagnetism]

Electric flux =

- (A) $E \cdot dA$
- (B) q/ϵ_0
- (C) Both
- (D) None

Q6. 1M [-0.33]

[Quantum Mechanics]

$|\psi|^2$ represents:

- (A) Energy
- (B) Probability density
- (C) Momentum
- (D) Force

Q7.

1M [-0.33]

[Thermal Physics]

Avg KE of molecule:

- (A) $(3/2)kT$
- (B) kT
- (C) $(1/2)kT$
- (D) $2kT$

Q8.

1M [-0.33]

[Optics]

Laser principle:

- (A) Spontaneous
- (B) Stimulated emission
- (C) Absorption
- (D) Reflection

Q9.

1M [no -ve]

[Solid State Physics]

Intrinsic semiconductor at 0K:

Enter your answer here

Q10.

1M [-0.33]

[Nuclear Physics]

Half-life formula:

- (A) $N = N_0 e^{(-\lambda \cdot B \cdot t)}$
- (B) $N = N_0/2$
- (C) $N_0 - N$
- (D) $N_0 e^{-\lambda t}$

Q11.

1M [-0.33]

[Solid State Physics]

Intrinsic semiconductor at 0K:

- (A) Conductor
- (B) Insulator
- (C) Superconductor
- (D) None

Q12.

1M [-0.33]

[Solid State Physics]

Intrinsic semiconductor at 0K:

- (A) Conductor
- (B) Insulator
- (C) Superconductor
- (D) None

Q13.

1M [-0.33]

[Thermal Physics]

Avg KE of molecule:

- (A) $(3/2)kT$
- (B) kT
- (C) $(1/2)kT$
- (D) $2kT$

Q14.

1M [-0.33]

[Classical Mechanics]

F=ma is:

- (A) 1st law
- (B) 2nd law
- (C) 3rd law
- (D) None

Q15.

1M [-0.33]

[General Aptitude] $5^2 + 12^2$:

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q16.

1M [-0.33]

[General Aptitude] $5^2 + 12^2$:

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q17.

1M [-0.33]

[Quantum Mechanics] $|\psi\rangle$ is:

- (A) Energy
- (B) Probability density
- (C) Momentum
- (D) Force

Q18.

1M [-0.33]

[General Aptitude] $5^2 + 12^2$:

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q19.

1M [-0.33]

[Solid State Physics]

Intrinsic semiconductor at 0K:

- (A) Conductor
- (B) Insulator
- (C) Superconductor
- (D) None

Q20.

1M [-0.33]

[General Aptitude] $5^2 + 12^2$:

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q21.

1M [-0.33]

[General Aptitude] $5^2 + 12^2$:

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q22.

1M [-0.33]

[General Aptitude] $5^2 + 12^2$:

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q23.

1M [-0.33]

[Thermal Physics]

Avg KE of molecule:

- (A) $(3/2)kT$
- (B) kT
- (C) $(1/2)kT$
- (D) $2kT$

Q24.

1M [-0.33]

[Solid State Physics]

Intrinsic semiconductor at 0K:

- (A) Conductor
- (B) Insulator
- (C) Superconductor
- (D) None

Q25.

1M [-0.33]

*[Engineering Mathematics]*d/dx of e^x :

- (A) e^x
- (B) xe^x
- (C) 1
- (D) $\ln(x)$

Q26.

1M [-0.33]

[Classical Mechanics]

Period of spring-mass:

- (A) $2\pi\sqrt{m/k}$
- (B) $2\pi\sqrt{k/m}$
- (C) m/k
- (D) k/m

Q27.

1M [-0.33]

[Nuclear Physics]

Half-life formula:

- (A) $N = N_0 e^{(-\lambda)t}$
- (B) $N = N_0/2$
- (C) $N_0 - N$
- (D) λt

Q28.

1M [-0.33]

[Quantum Mechanics] $|\psi|^2$ represents:

- (A) Energy
- (B) Probability density
- (C) Momentum
- (D) Force

Q29.

1M [no -ve]

[Classical Mechanics]

Period of spring-mass:

Enter your answer here

Q30.

1M [no -ve]

*[Engineering Mathematics]*d/dx of e^x :

Enter your answer here

Q31.

2M [-0.66]

[Classical Mechanics] $F=ma$ is:

- (A) 1st law
- (B) 2nd law
- (C) 3rd law
- (D) None

Q32.

2M [-0.66]

[Classical Mechanics]

F=ma is:

- (A) 1st law
- (B) 2nd law
- (C) 3rd law
- (D) None

Q33.

2M [-0.66]

[Solid State Physics]

Intrinsic semiconductor at 0K:

- (A) Conductor
- (B) Insulator
- (C) Superconductor
- (D) None

Q34.

2M [no -ve]

[General Aptitude] $5^2 + 12^2$:

*Enter your answer here***Q35.**

2M [-0.66]

[Thermal Physics]

Avg KE of molecule:

- (A) $(3/2)kT$
- (B) kT
- (C) $(1/2)kT$
- (D) $2kT$

Q36.

2M [-0.66]

[Solid State Physics]

Intrinsic semiconductor at 0K:

- (A) Conductor
- (B) Insulator
- (C) Superconductor
- (D) None

Q37.

2M [no -ve]

[Thermal Physics]

Avg KE of molecule:

*Enter your answer here***Q38.**

2M [-0.66]

[General Aptitude] $5^2 + 12^2$:

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q39.

2M [no -ve]

[Optics]

Laser principle:

Enter your answer here

Q40.

2M [-0.66]

[Engineering Mathematics] d/dx of e^x :

- (A) e^x
- (B) xe^x
- (C) 1
- (D) $\ln(x)$

Q41.

2M [-0.66]

[Classical Mechanics] $F=ma$ is:

- (A) 1st law
- (B) 2nd law
- (C) 3rd law
- (D) None

Q42.

2M [no -ve]

[Classical Mechanics] $F=ma$ is:

Enter your answer here

Q43.

2M [no -ve]

[Nuclear Physics]

Half-life formula:

Enter your answer here

Q44.

2M [-0.66]

[Solid State Physics]

Intrinsic semiconductor at 0K:

- (A) Conductor
- (B) Insulator
- (C) Superconductor
- (D) None

Q45.

2M [no -ve]

[Nuclear Physics]

Half-life formula:

- (A) $N = N_0 e^{(-\lambda \cdot B \cdot t)}$
- (B) $N = N_0/2$
- (C) $N_0 - N$
- (D) $N = N_0 e^{(-\lambda t)}$

Q46.

2M [-0.66]

[General Aptitude]

$5^2 + 12^2:$

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q47.

2M [-0.66]

[Quantum Mechanics] $|\psi\rangle$ is:

- (A) Energy
- (B) Probability density
- (C) Momentum
- (D) Force

Q48.

2M [-0.66]

[Quantum Mechanics] $|\psi\rangle$ is:

- (A) Energy
- (B) Probability density
- (C) Momentum
- (D) Force

Q49.

2M [-0.66]

[Solid State Physics]

Intrinsic semiconductor at 0K:

- (A) Conductor
- (B) Insulator
- (C) Superconductor
- (D) None

Q50.

2M [-0.66]

[Classical Mechanics]

Period of spring-mass:

- (A) $2\pi\sqrt{m/k}$
- (B) $2\pi\sqrt{k/m}$
- (C) m/k
- (D) k/m

Q51.

2M [-0.66]

[General Aptitude]

$5^2 + 12^2:$

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q52.

2M [-0.66]

[Solid State Physics]

Intrinsic semiconductor at 0K:

- (A) Conductor
- (B) Insulator
- (C) Superconductor
- (D) None

Q53.

2M [-0.66]

[Engineering Mathematics] d/dx of e^x :

- (A) e^x
- (B) xe^x
- (C) 1
- (D) $\ln(x)$

Q54.

2M [-0.66]

[General Aptitude] $5^2 + 12^2$:

- (A) 13
- (B) 169
- (C) 17
- (D) 144

Q55.

2M [no -ve]

[Engineering Mathematics] d/dx of e^x :

*Enter your answer here***Q56.**

2M [no -ve]

[Nuclear Physics]

Half-life formula:

*Enter your answer here***Q57.**

2M [-0.66]

[Classical Mechanics] $F=ma$ is:

- (A) 1st law
- (B) 2nd law
- (C) 3rd law
- (D) None

Q58.

2M [-0.66]

[Optics]

Laser principle:

- (A) Spontaneous
- (B) Stimulated emission
- (C) Absorption
- (D) Reflection

Q59.

2M [no -ve]

[Optics]

Laser principle:

Enter your answer here

Q60.

2M [no -ve]

[Engineering Mathematics] d/dx of e^x :

Enter your answer here

Q61.

2M [no -ve]

[Quantum Mechanics] $|\psi\rangle$ is:

Enter your answer here

Q62.

2M [no -ve]

[Electromagnetism]

Electric flux =

Enter your answer here

Q63.

2M [-0.66]

[Classical Mechanics] $F=ma$ is:

- (A) 1st law
- (B) 2nd law
- (C) 3rd law
- (D) None

Q64.

2M [-0.66]

[Quantum Mechanics] $|\psi\rangle$ is:

- (A) Energy
- (B) Probability density
- (C) Momentum
- (D) Force

Q65.

2M [no -ve]

[General Aptitude] $5^2 + 12^2$:

Enter your answer here